

NAME : SIVAKRISHNA

REG NO:1924260046

Set 18: Product Inventory Management

Data Visualization using R

Aim

To analyse product inventory using a bar chart, a stacked bar chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the inventory data as a data frame. The data has no category column, so categories are assumed.
3. Question 1: draw the bar chart of quantity with `geom_col()`.
4. Question 2: draw the stacked bar chart of quantity by category with `geom_col()`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset (Category is assumed, as the data has no category column) inv <-
data.frame(Product_ID = 1:3, Product = c("Product A", "Product B", "Product C"),
           Quantity = c(250, 175, 300), Price = c(20, 15, 18),
           Category = c("Category 1", "Category 2", "Category 1")) print(inv)
print(aggregate(Quantity ~ Category, inv, sum))

# 1. Bar chart - quantity of each product
p1 <- ggplot(inv, aes(Product, Quantity)) + geom_col(fill = "steelblue") +
geom_text(aes(label = Quantity), vjust = -0.5) + labs(title = "Quantity of Each
Product", x = "Product", y = "Quantity Available")

# 2. Stacked bar chart - products within categories p2 <- ggplot(inv,
aes(Category, Quantity, fill = Product)) + geom_col() + labs(title =
"Quantity by Category", x = "Category", y = "Quantity")

# 3. Table tb <- tableGrob(inv[, 1:4],
rows = NULL)

# 4. Dashboard dash <- arrangeGrob(p1, p2, tb, layout matrix = rbind(c(1,2), c(3,3)), top =
"Inventory Dashboard")

ggsave("set18_q1.png", p1, width = 6, height = 4, bg = "white") ggsave("set18_q2.png",
p2, width = 6, height = 4, bg = "white")
ggsave("set18_q3.png", arrangeGrob(tb, top = "Product Inventory Table"), width = 5, height = 2, bg = "white")
ggsave("set18_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

	Product_ID	Product	Quantity	Price	Category
1	1	Product A	250	20	Category 1

2	2 Product B	175	15 Category 2
3	3 Product C	300	18 Category 1
Category Quantity			
1	Category 1	550	
2	Category 2	175	

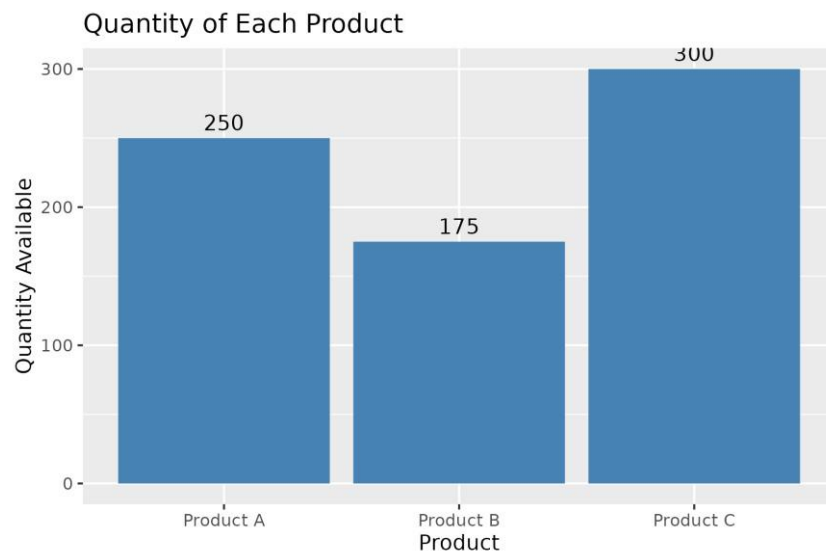


Figure 18.1: Q1 - Quantity of each product

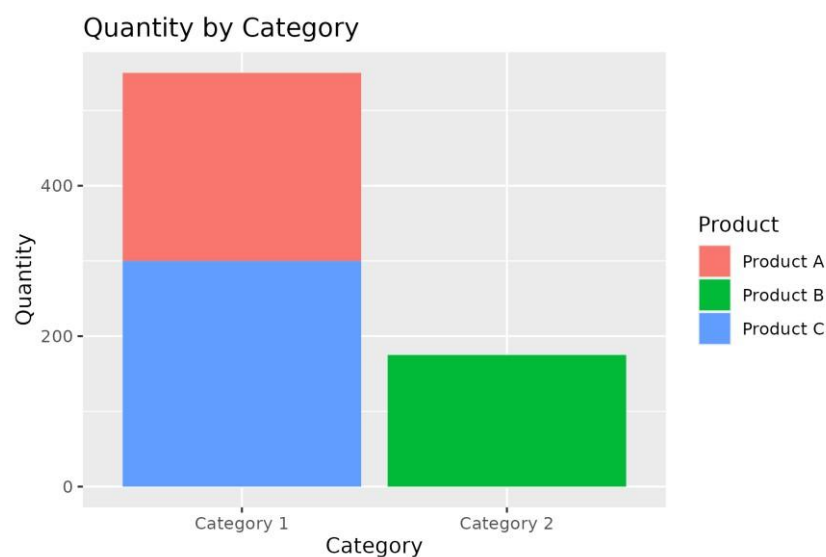


Figure 18.2: Q2 - Stacked bar chart by category (assumed)

Product Inventory Table

Product_ID	Product	Quantity	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Figure 18.3: Q3 - Inventory table

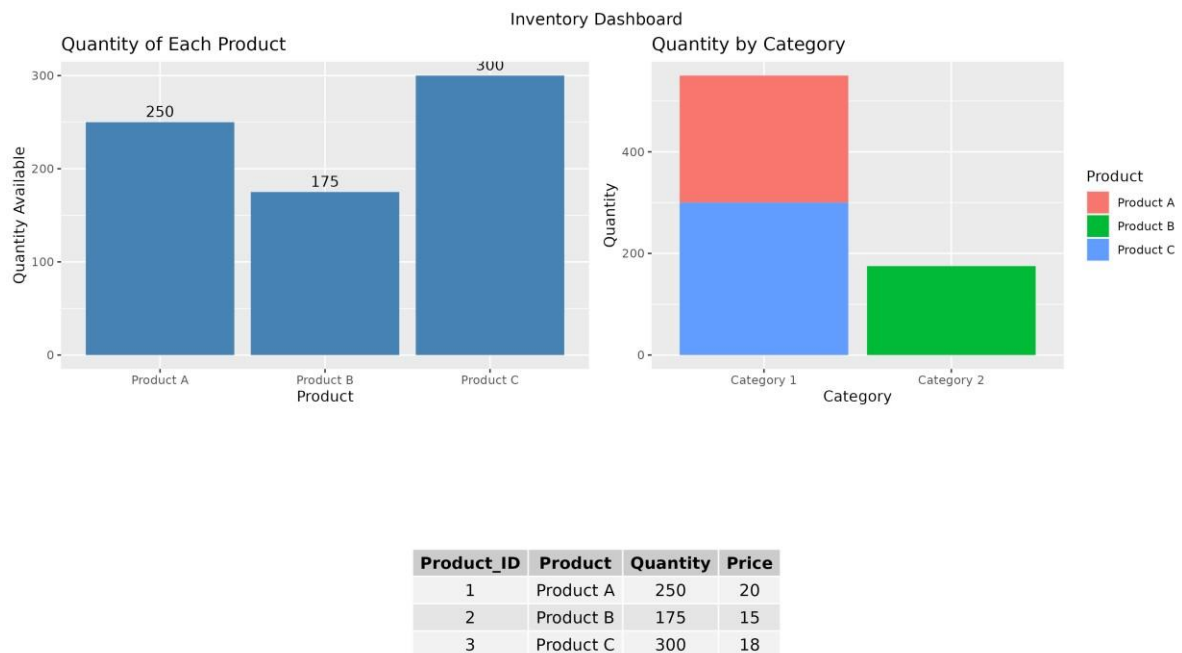


Figure 18.4: Q4 - Dashboard

Result

- Product C has the most stock (300) and Product B the least (175).
- Total stock is 725 units.
- Categories are assumed, so the stacked chart is for demonstration.

Set 19: Survey Responses Analysis

Data Visualization using R

Aim

To analyse survey responses using a grouped bar chart, a stacked bar chart, a table and a dashboard in R.

Procedure

8. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
9. Create the survey data as a data frame.
10. Question 1: draw the grouped bar chart of Question 1 answers with `geom_bar(position = "dodge")`.
11. Question 2: draw the stacked bar chart of all three questions with `geom_bar()`.
12. Question 3: make the table with `tableGrob()`.
13. Question 4: join the two charts and the table into a dashboard with `arrangeGrob()`.
14. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset survey <- data.frame(Survey_ID = 1:3, Question1 =
c("A","B","C"), Question2 = c("B","A","A"), Question3 =
c("C","D","B")) print(survey)
long <- stack(survey[, 2:4]); names(long) <- c("Answer", "Question") print(table(long$Question,
long$Answer))

# 1. Grouped bar chart - Question 1
p1 <- ggplot(survey, aes(Question1, fill = factor(Survey_ID))) + geom_bar(position
= "dodge") + labs(title = "Answers of Question 1", x = "Answer", y = "Count", fill
= "Survey_ID")

# 2. Stacked bar chart - all questions
p2 <- ggplot(long, aes(Question, fill = Answer)) + geom_bar() +
labs(title = "Responses of All Questions", x = "Question", y = "Count")

# 3. Table tb <- tableGrob(survey, rows
= NULL)

# 4. Dashboard dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top =
"Survey Dashboard")

ggsave("set19_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set19_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set19_q3.png", arrangeGrob(tb, top = "Survey Response Table"), width = 5, height = 2, bg = "white")
ggsave("set19_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

	Survey_ID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

	A	B	C	D
Question1	1	1	1	0
Question2	2	1	0	0
Question3	0	1	1	1

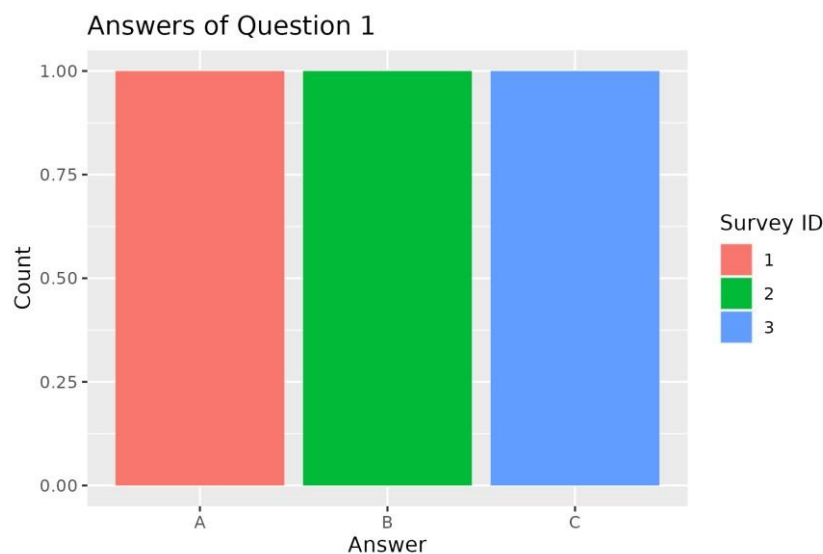


Figure 19.1: Q1 - Grouped bar chart (Question 1)

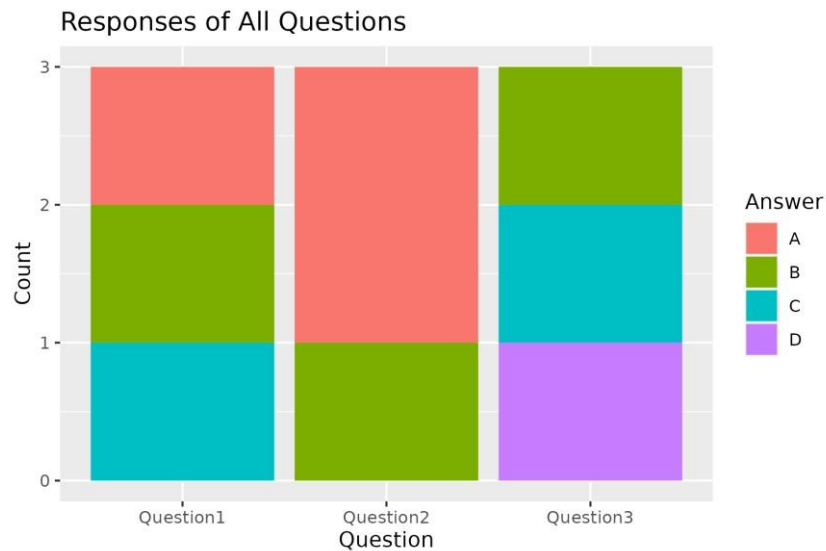
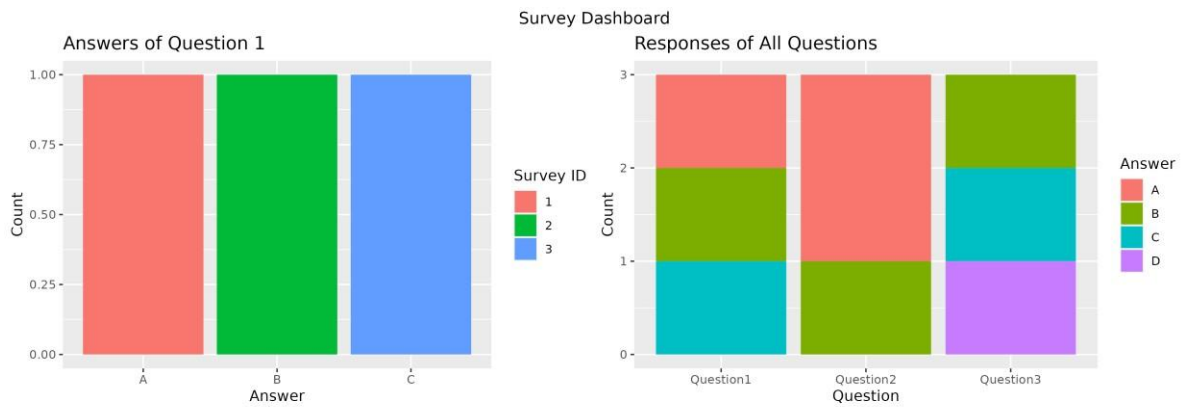


Figure 19.2: Q2 - Stacked bar chart (all questions)

Survey Response Table

Survey_ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 19.3: Q3 - Survey response table



Survey_ID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

Figure 19.4: Q4 - Dashboard

Result

- In Question 1, answers A, B and C were chosen once each.

- Question 2 has A twice and B once; Question 3 has B, C and D once each.
- All four parts were completed successfully in R.

Set 20: Stock Analysis

Data Visualization using R

Aim

To analyse stock prices using a line chart, a bar chart, a table and a dashboard in R.

Procedure

15. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
16. Create the stock data as a data frame.
17. Question 1: draw the line chart of the three stocks with `geom_line()`.
18. Question 2: calculate the daily % change of Stock A with `diff()` and draw the bar chart.
19. Question 3: make the table with `tableGrob()`.
20. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
21. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset stocks <- data.frame(Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
#                               Stock_A = c(100, 105, 110), Stock_B = c(150, 152, 148),
#                               Stock_C = c(120, 118, 122))
print(stocks) change <- data.frame(Date = stocks$Date[-1],
#                                   Change = diff(stocks$Stock_A) / head(stocks$Stock_A, -1) * 100) print(change)

# 1. Line chart - three stocks p1 <- ggplot(stocks,
# aes(Date)) + geom_line(aes(y = Stock_A, colour = "Stock A")) + geom_line(aes(y = Stock_B, colour = "Stock B")) +
# geom_line(aes(y = Stock_C, colour = "Stock C")) + labs(title = "Stock Prices", x = "Date", y = "Price", colour = "Stock")

# 2. Bar chart - daily % change of Stock A
p2 <- ggplot(change, aes(factor(Date), Change)) + geom_col(fill = "seagreen") +
geom_text(aes(label = paste0(round(Change, 2), "%")), vjust = -0.5) + labs(title = "Daily % Change of Stock A", x = "Date", y = "Change (%)")

# 3. Table tb <- tableGrob(stocks,
# rows = NULL)

# 4. Dashboard dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Stock Dashboard")

ggsave("set20_q1.png", p1, width = 6, height = 4, bg = "white") ggsave("set20_q2.png",
p2, width = 6, height = 4, bg = "white")
ggsave("set20_q3.png", arrangeGrob(tb, top = "Stock Price Table"), width = 5, height = 2, bg = "white")
ggsave("set20_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

```
  Date Stock_A Stock_B Stock_C
1 2023-01-01   100   150   120
```

2	2023-01-02	105	152	118	3	2023-01-03	110	148	122
Date Change									
1	2023-01-02	5.000000							
2	2023-01-03	4.761905							

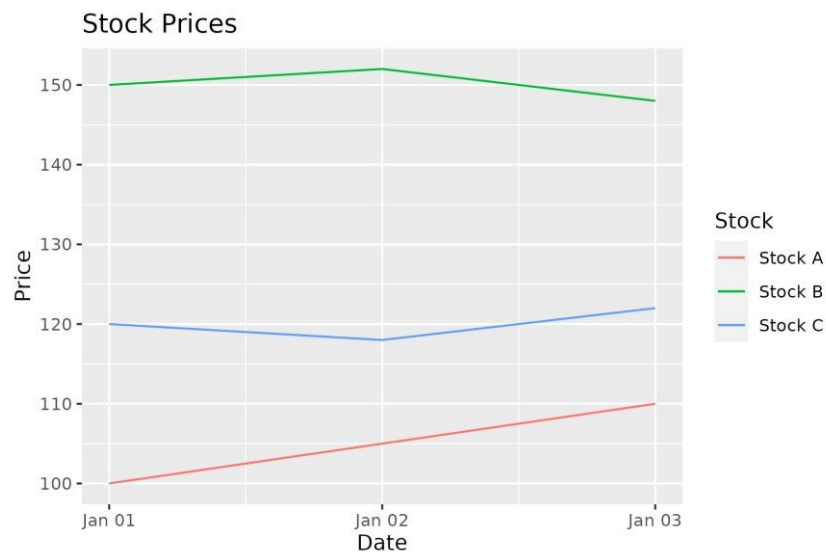


Figure 20.1: Q1 - Stock price line chart

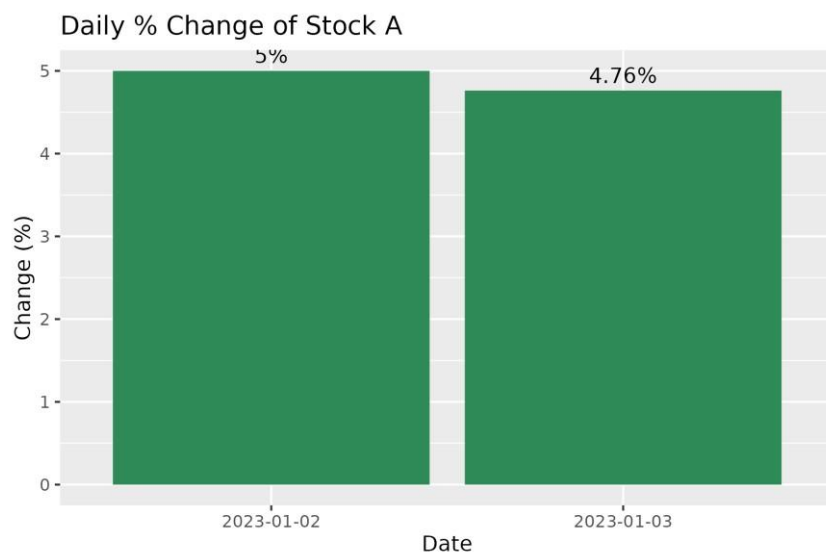


Figure 20.2: Q2 - Daily % change of Stock A

Stock Price Table

Date	Stock_A	Stock_B	Stock_C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

Figure 20.3: Q3 - Stock price table

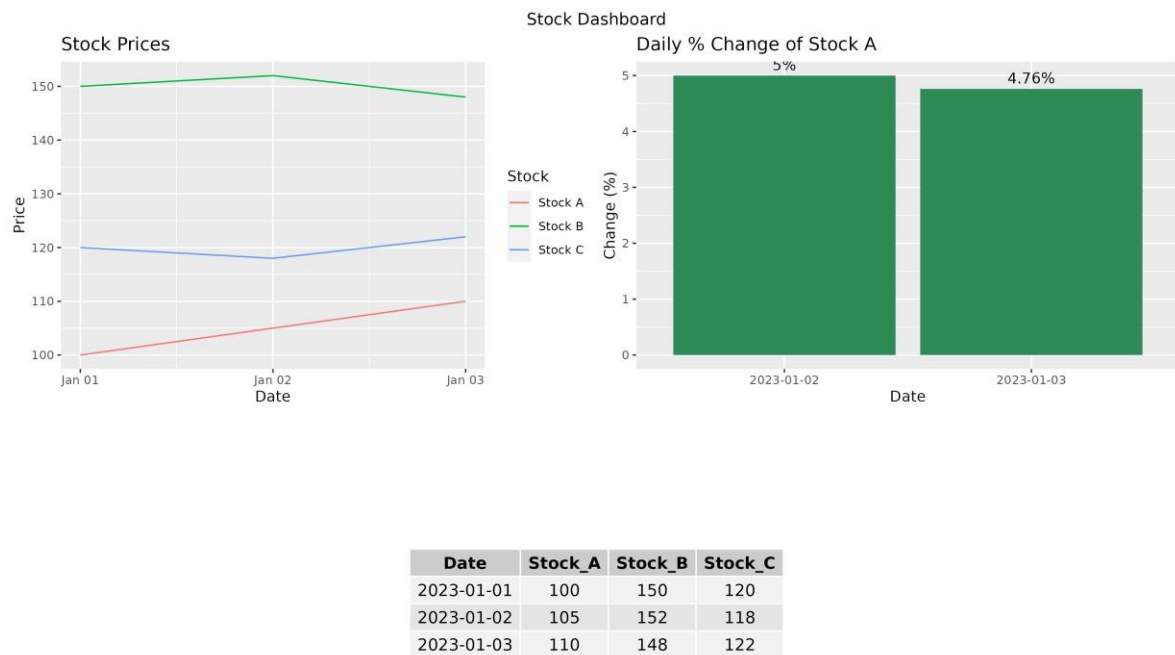


Figure 20.4: Q4 - Dashboard

Result

- Stock A rose from 100 to 110, with daily changes of +5.00% and +4.76%.
- Stock B and Stock C moved up and down.
- All four parts were completed successfully in R.

Set 21: Energy Consumption Analysis

Data Visualization using R

Aim

To analyse energy consumption data in R using a histogram, a density plot, a bubble scatter plot, a bar chart and a dashboard with filters.

Procedure

22. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
23. Save the data as `Energy_Consumption_Data.csv` and import it with `read.csv()`.
24. Question 1: plot the histogram with `geom_histogram()` and the density curve with `geom_density()`.
25. Question 2: draw the scatter plot of temperature against units with bubble size = `Peak_Hours` and alpha for transparency.
26. Question 3: find the average `Renewable_Usage` of each sector with `aggregate()` and draw the bar chart.
27. Question 4: draw the monthly line chart and the Sector vs Region heatmap, join them into a dashboard, and apply the Month and Sector filters with `subset()`.
28. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra) options(width
= 95)

# Create the CSV file and import it
data.frame(
  Sector = c("Residential", "Commercial", "Industrial", "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
) -> d
write.csv(d, "Energy_Consumption_Data.csv", row.names = FALSE)
energy <- read.csv("Energy_Consumption_Data.csv") energy$Month
<- factor(energy$Month, levels = c("Jan", "Feb", "Mar")) print(energy)

# 1. Histogram and density plot
print(aggregate(Units_Consumed ~ Sector, energy, mean)) p1 <- ggplot(energy,
aes(Units_Consumed)) + geom_histogram(aes(y = after_stat(density)), binwidth = 150, fill =
"skyblue", colour = "white") + geom_density(colour = "red") +
labs(title = "Histogram and Density of Units Consumed", x = "Units Consumed (kWh)", y = "Density")

# 2. Scatter plot with bubble size and
transparency
cat("Correlation:", cor(energy$Temperature, energy$Units_Consumed), "\n") p2 <- ggplot(energy,
aes(Temperature, Units_Consumed, size = Peak_Hours)) + geom_point(alpha = 0.5, colour = "blue")
+ labs(title = "Temperature vs Units Consumed", x = "Temperature (C)", y = "Units Consumed
(kWh)")

# 3. Average renewable usage by sector
avg <- aggregate(Renewable_Usage ~ Sector, energy, mean) print(avg)

p3 <- ggplot(avg, aes(Sector, Renewable_Usage, fill = Sector)) + geom_col(show.legend = FALSE) +
geom_text(aes(label = paste0(Renewable_Usage, "%")), vjust = -0.5) + labs(title = "Average
Renewable Usage by Sector", y = "Renewable Usage (%)")

# 4. Dashboard (line chart + heatmap); filters are applied with subset() dash <- function(d, title) { line <-
ggplot(aggregate(Units_Consumed ~ Month, d, sum), aes(Month, Units_Consumed, group = 1)) +
geom_line() + geom_point(size = 3) + labs(title = "Monthly Energy Usage", y = "Units (kWh)") heat <-
ggplot(d, aes(Region, Sector, fill = Units_Consumed)) + geom_tile() + geom_text(aes(label =
Units_Consumed)) + scale_fill_gradient(low = "yellow", high = "red") + labs(title = "Heatmap: Sector
vs Region", fill = "Units") arrangeGrob(line, heat, ncol = 2, top = title)
}
d1 <- dash(energy, "Energy Dashboard (No Filter)") d2 <- dash(subset(energy,
Month %in% c("Jan", "Feb") & Sector != "Industrial"), "Energy
Dashboard (Filter: Jan-Feb, Residential and Commercial)")

ggsave("set21_q1.png", p1, width = 6, height = 4, bg = "white") ggsave("set21_q2.png",
p2, width = 6, height = 4, bg = "white") ggsave("set21_q3.png", p3, width = 6, height =
4, bg = "white") ggsave("set21_q4a.png", d1, width = 12, height = 5, bg = "white")
ggsave("set21_q4b.png", d2, width = 12, height = 5, bg = "white")
```

Output

	Sector	Region	Month	Temperature	Units_Consumed	Cost	Renewable_Usage	Peak_Hours
1	Residential	North	Jan	15	320	2100	22	4
2	Commercial	South	Jan	24	540	3600	18	6
3	Industrial	West	Feb	20	880	5900	12	8
4	Residential	East	Feb	18	350	2300	25	5
5	Commercial	North	Mar	28	610	4100	20	7
6	Industrial	South	Mar	30	920	6200	15	9
Sector Units_Consumed								
1	Commercial				575			
2	Industrial				900			
3	Residential				335			
Correlation: 0.654655								
Sector Renewable_Usage								
1	Commercial					19.0		
2	Industrial					13.5		
3	Residential					23.5		

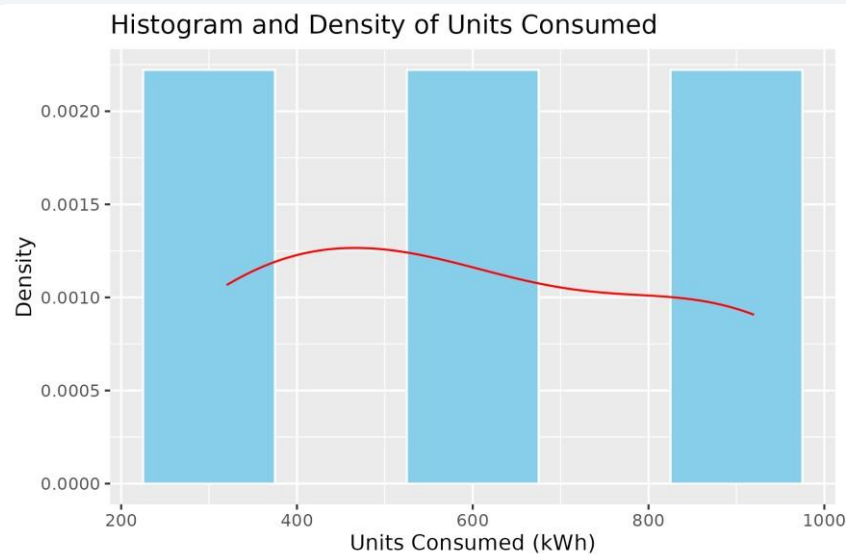


Figure 21.1: Q1 - Histogram and density plot

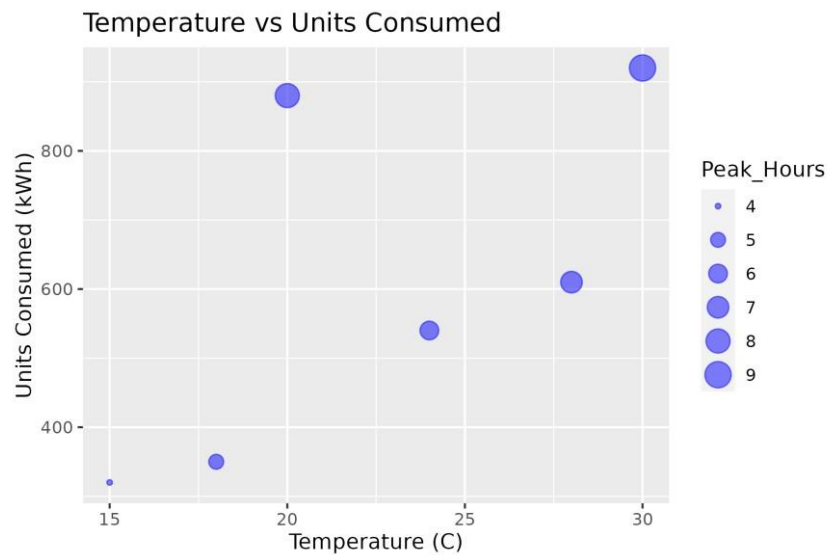


Figure 21.2: Q2 - Temperature vs units (bubble = peak hours)

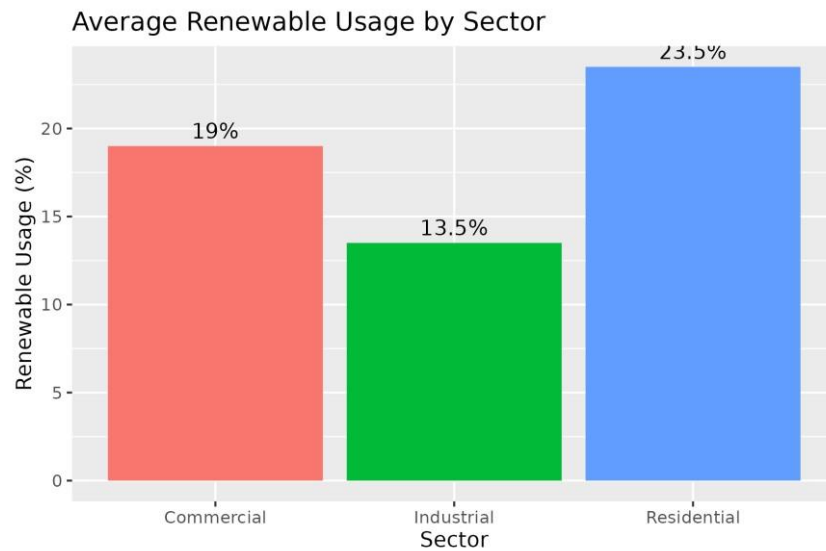


Figure 21.3: Q3 - Average renewable usage by sector

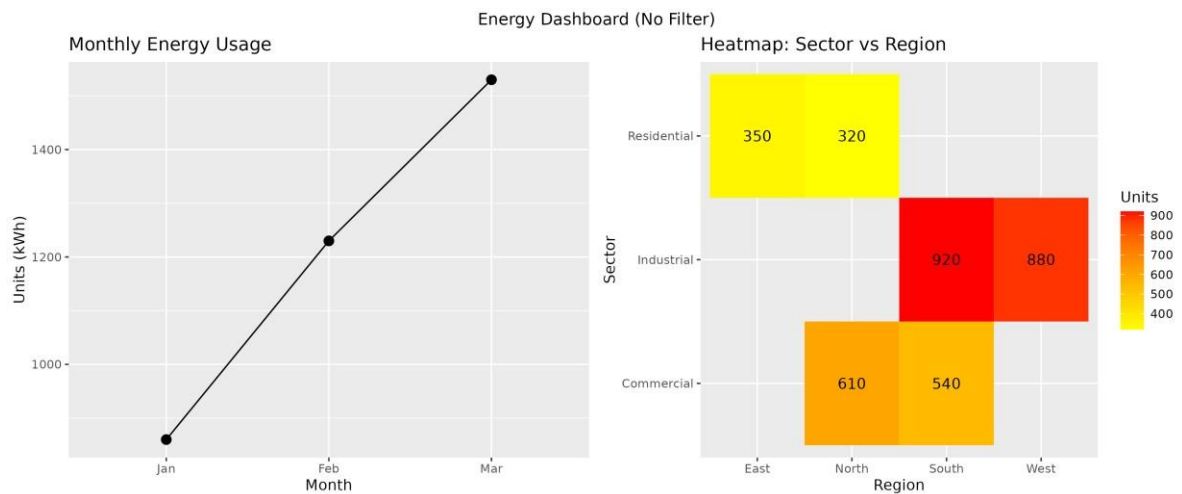


Figure 21.4: Q4 - Dashboard (no filter)

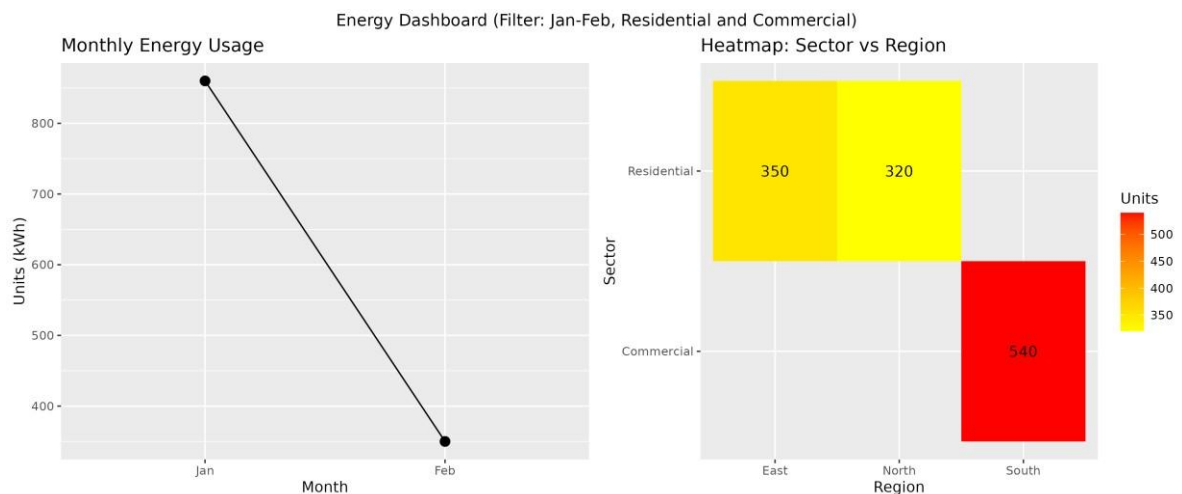


Figure 21.5: Q4 - Dashboard (filter: Jan-Feb, Residential and Commercial)

Result

- Industrial uses the most energy (average 900 kWh) and Residential the least (335 kWh).
- Temperature and units have a positive correlation of 0.65.

- Residential has the highest renewable usage (23.5%) and Industrial the lowest (13.5%), so Industrial has the most scope to improve.
- Energy use rises from January to March, and the heatmap shows the highest use in Industrial.